Jev in a Multi-Step Workflow: Decisions Predict Failures, Not Which Reviews Pay

Anonymous ACL submission

Abstract

Typed decision models such as Jev return a label with a numeric confidence, and their primitives have been benchmarked on single tasks. We ask what single-task benchmarks leave open: when a workflow chains heterogeneous typed decisions, which signals identify its failures, and can they identify the costly errors for which human review is worthwhile? We chain Jev's categorical, binary and ordinal primitives into a dependent bug-triage workflow and evaluate it on Mozilla and Linux kernel bugs. In pre-registered tests on 500 new bugs, a frozen estimator built on what the workflow decided ranked wrong actions better than final-step confidence (AUROC 0.76 vs. 0.55). The gap replicated across five models from four sources, and Jev's estimator transferred unchanged to an independently trained Jev-style model; the signal was mostly the chosen action's historical error rate. Yet under the pre-registered costs it did not yield a cost-effective review policy: at a 10% review budget no strategy cost less than no review, though an exploratory sensitivity analysis shows this reverses for three of five models when wrong accepts cost twice as much. An oracle shows the expensive errors exist, but the signals tested could not isolate them, and a case-level score failed a pre-registered feasibility check (power at most 0.20), so the confirmatory study was stopped before new data. Predicting which decisions are wrong is not knowing which are worth correcting.

1 Introduction

A model can return a valid, well-typed decision and still be wrong. When several such decisions are chained into a workflow, errors propagate, and a common hope is that the steps' reported confidence can say when to hand a case to a human. Typed decision models make this hope concrete. Jev, TypeSafe's decision API, answers closed questions through three primitives: a categorical `choice`, a binary `noul` and an ordinal `score`, each with a number between 0 and 1. Prior work has benchmarked these primitives and their calibration on single requests (Deußer et al., 2026; Meng, 2026; Denney and DiGiuseppe, 2026), and argued in general that error risk is not review value (Park et al., 2026).

We study an operational question that single-request benchmarks leave open. *When a workflow chains heterogeneous typed decisions, which signals identify the cases where it fails, and can those signals identify the specific costly errors for which human review is worthwhile?* The two halves can come apart: a signal can rank failures well and still not say which reviews pay.

We chain Jev's three primitives into a dependent bug-triage workflow (route, assess regression, rate severity) and study it on 202 Mozilla bugs labeled by triage and 240 Linux kernel bugs labeled by their reporters, followed by pre-registered tests on 500 new bugs and four further models. Our contributions:

1. **Workflow failure prediction** (Sections 5 and 6). A frozen decision-feature estimator ranks wrong workflow actions better than final-step confidence on 500 new bugs, and the gap replicates across five models from four sources. Most of the signal is the chosen action's historical error rate.

2. **Prediction is not intervention utility** (Section 7). At a fixed review budget and the pre-registered costs, that signal does not reduce cost, though an exploratory sensitivity analysis shows that whether review pays depends on the cost ratio; an oracle locates the value in wrongly accepted cases that the signals cannot isolate, and a case-level score fails a pre-registered feasibility check, so the confirmatory study is stopped by rule before new data.

3. **Supporting evidence** (Section 4). Inside the workflow the three primitives' confidence signals behave differently, thresholds set on constructed data do not transfer, and `score` compresses severity on both trackers. These results agree with the single-request audits and explain the workflow-level findings.

All data collection and confirmatory analyses were pre-registered: each protocol was committed, with its SHA-256 hash, before the data it governs existed. The anonymized supplementary archive lists the commit dates and hashes, and deviations are disclosed (Appendix D).

2 Related work

Evaluations of Jev and open decision models. Jev has been benchmarked zero-shot on 37 datasets, where its `choice` probabilities were well calibrated and its binary probabilities ranked well but sat poorly relative to a 0.5 threshold (Deußer et al., 2026). A calibration audit (described by its author as exploratory) found that calibration changed with the question form, that 71.6% of four-class answers sat at exactly 1.00, leaving a top-class threshold only 53 operating points, and that no threshold certified a 5% automatic tier on the four-class task (Meng, 2026). On political-science replications Jev's calibration advantage was not consistent (Denney and DiGiuseppe, 2026), and an early evidence audit found no independent accuracy advantage yet for the typed readout (Tang and Zheng, 2026). Open instruction-tuned models give Jev-like decisions by reading label logits in one forward pass (Geuter et al., 2026), the readout we use for the general models. These studies evaluate single requests; none studies `score` inside a workflow, composition of dependent decisions, or the cost of human review.

Review value and deferral. Learning to defer (Madras et al., 2018), selective prediction (Geifman and El-Yaniv, 2017), cost-sensitive deferral under workload limits (Alves et al., 2024) and cascades with escalation (Jung et al., 2025; Bouchard, 2026) address deferral of single decisions. Closest to our intervention result, Park et al. (2026) argue that review priority should depend on wrongness, repairability, impact and cost rather than risk alone, and show on a QA benchmark that review-value ranking leaves less residual error than risk ranking. Our result is an instance of that distinction in a dependent typed-decision workflow, with a frozen cost table, pre-registered tests and a feasibility check for a case-level policy.

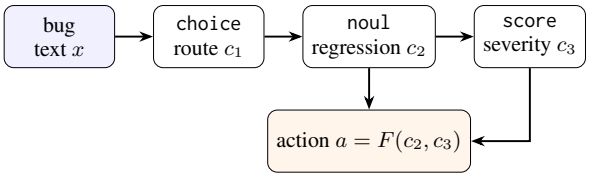


Figure 1: The workflow. Each node reads the bug text plus all upstream decisions and is called three times. The action rule F uses c_2 and c_3; a review replaces the action with *escalate*.

Composition and trajectories. Errors compound across pipeline stages, and independently calibrated stages lack joint guarantees (Kotte, 2026; Chowdhury et al., 2019); step-level calibration need not imply trajectory-level calibration for LLM agents (Lin et al., 2026); and the product rule for combining confidences assumes conditional independence (Kittler et al., 1998). LLM raters compress ordinal scores toward the middle (Zhang et al., 2026).

Positioning. We are not aware of prior work that tests, in a dependent workflow of structurally different typed decisions with real labels, held-out data and several models, which signals predict workflow failure and whether they identify the reviews worth paying for. We make no priority claim: our search was web-based, and the cited Jev studies appeared in September and October 2026.

3 Setting

Workflow. Node 1 (`choice`) routes a bug to the network or storage subsystem; node 2 (`noul`) assesses whether it is a regression; node 3 (`score`) rates severity on three levels. Each node reads the bug text plus the upstream decisions, appended by a fixed adapter (Fig. 1), and is called three times; the answers are reduced to the modal decision (mean probability for `noul`). All calls used `jev-1.13.0`.

Target action and cost. The action rule $F(c_2, c_3)$ maps an unclear regression answer to *request evidence*, severity 2 to *escalate*, severity 1 with a regression to *defer*, and everything else to *accept* (Table 6). The target is $a^* = F$ applied to the reference labels. It is a construct we defined, and every strategy result is conditional on it. Costs come from a frozen matrix (Table 7): a wrong *accept* costs 6–10, any other mismatch 1, and each

	Mozilla (triage)	Kernel (reporter)
`choice` at ≥ 0.90	85%	96%
`choice` conf. AUROC	0.90 [0.83, 0.96]	0.98 [0.95, 1.00]
`noul` value AUROC	0.53 [0.42, 0.65]	0.41 [0.32, 0.50]
`score` conf. AUROC	0.58 [0.50, 0.66]	0.57 [0.50, 0.64]
Agreement: `choice`	0.89 [0.83, 0.92]	0.975 [0.95, 0.99]
`noul`	0.80 [0.74, 0.85]	0.725 [0.67, 0.78]
`score`	0.53 [0.46, 0.60]	0.44 [0.38, 0.50]

Table 1: Each primitive's confidence signal, as AUROC for predicting the node's own agreement with the reference label (95% bootstrap CI), and node agreement. Full table and reliability diagrams in Appendix B.

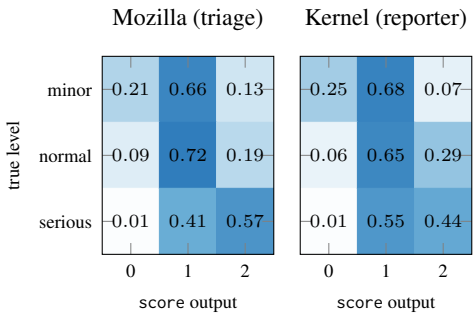


Figure 2: Row-normalized confusion of `score` against reference severity (repeat-unstable outputs excluded). On both trackers the middle column absorbs mass from both extremes: minor bugs are rated up, serious bugs down.

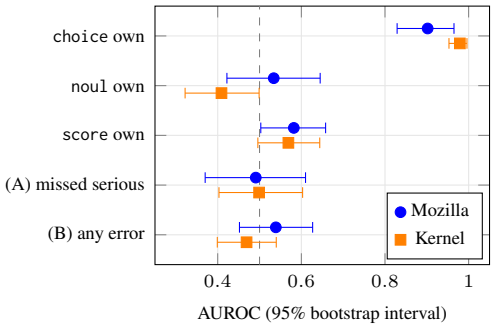


Figure 3: Development-set AUROCs with 95% intervals; dashed line: chance. Only `choice`'s confidence discriminates its own errors. Composed confidence is at chance for both escalation targets, (A) and (B).

escalation an extra 1 for human time. A review turns a case's action into *escalate*, so reviewing a wrongly accepted case saves 6 or 9, and any other review loses 1–2.

Data. *Mozilla:* 202 fixed Firefox networking and storage bugs; severity is assigned by triage and the regression flag is a keyword. *Linux kernel:* 240 fixed networking and storage bugs; severity is set by the reporter and the regression flag is a field. Both development samples are balanced across (subsystem, level) cells. The pre-registered tests use 500 *new* bugs, 250 per tracker, sampled at each tracker's natural mix. Agreement is with reference labels, not ground truth.

Statistics. AUROCs use the Mann–Whitney estimator with percentile bootstrap intervals over bugs ($B = 2{,}000$, stratified by tracker). Confirmatory tests are one-sided paired bootstrap tests with Holm correction within each pre-registered family.

4 Supporting evidence: the primitives inside the workflow

The three primitives' signals are not one kind of quantity (Table 1). `choice` confidence is saturated (85–96% of bugs at ≥ 0.90) yet discriminative, because its rare low-confidence answers are where its errors are. `noul`'s confidence-like value $2|\bar{p} - 0.5|$ carries no usable information about its own correctness. `score`'s confidence is flat across true severity levels, so it gives no warning of a systematic bias (Fig. 2): on both trackers most minor bugs are rated higher than their label (79% [0.67, 0.88] and 75% [0.65, 0.83]) and many serious bugs lower (43% and 56%), a central-tendency pattern reported for LLM raters elsewhere (Zhang et al., 2026). These findings agree with the single-request audits where they overlap: `choice` is saturated yet discriminative, and thresholds must be validated on target data. On `noul`, the audits report accurate, recalibratable yes/no answers on sentiment, whereas on our regression question its probability did not discriminate its own correctness, so binary-probability quality looks task-dependent.

Fixed thresholds do not transfer. Two escalation thresholds designed on constructed data lost their selectivity on real text: a `choice` trigger fired on 85–96% of bugs everywhere, and a confidence-product cut that escalated 73% of constructed items (target 67%) escalated 88% and 78% of real bugs (targets 41–42%), because real text pulls `noul` toward 0.5.

Composed confidence does not pick escalations (exploratory). Escalating, on top of the decision-based action, the bugs with the lowest composed confidence did no better than random selection on either tracker. Composed confidence ranked missed serious bugs at AUROC 0.49 [0.37, 0.61] and 0.50 [0.40, 0.60], and any decision error

at 0.54 [0.45, 0.63] and 0.47 [0.40, 0.54] (Fig. 3). Only `choice`'s confidence discriminates, and it belongs to the one node the action rule does not use. The obvious comparison, decision-based against confidence-based composition, is confounded, because the target is itself a function of the decisions. These analyses were specified after the Mozilla results and motivated the pre-registered tests below (Appendix C).

5 A pre-registered held-out test

We trained one decision-feature estimator (E_{dec}: L_2-regularized logistic regression on the `score` level, the `noul` decision and the workflow action) on all 442 development bugs and froze it by hash before any new data existed. We then collected the 500 new bugs (4,500 Jev calls). Pre-registered hypotheses: HR1, E_{dec} ranks wrong workflow actions better than low final-step confidence; HR2, it ranks above chance on each tracker.

Both are supported. HR1: E_{dec} reaches AUROC 0.763 against 0.549 for final-step confidence, a difference of $+0.214$ (one-sided 95% lower bound $+0.158$). HR2: AUROC is 0.811 on Mozilla (lower bound 0.761) and 0.697 on the kernel (lower bound 0.643). The estimator's predicted failure rate matches the observed rate overall, but it outputs only 7–8 distinct risk levels (ECE 0.18–0.20), so it is a ranking, not a calibrated probability. Held-out and development bugs share no bug IDs, no exact texts and no near-duplicates (word 5-gram Jaccard ≥ 0.8; the highest similarity of any held-out bug to a development bug is 0.58).

6 Cross-model replication

We ran the same 942 bugs, questions and adapter through four further models. **OpenJev 27B** (OpenJev, 2026) is an independently trained Jev-style decision model (Hugging Face `openjev/openjev`, commit `b4a89f5`; unrelated projects share the name), served on one H100 exactly as its authors specify. **Qwen2.5-7B** (Qwen et al., 2024), **OLMo-2-7B** (Team OLMo et al., 2024) and **Phi-3.5-mini** (Abdin et al., 2024) run behind a Jev-compatible readout: next-token probabilities over the allowed labels, with TypeSafe's documented confidence formulas. Each model's estimator was fitted on its own outputs for the 442 development bugs and tested on the 500 held-out bugs; the tests mirror HR1 and HR2, Holm-corrected across all 8. Three

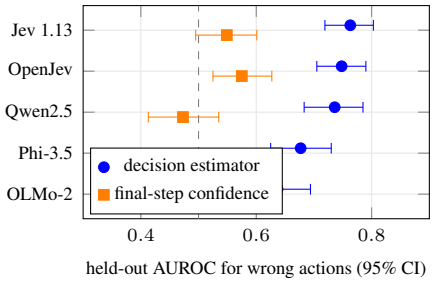


Figure 4: Across five models, the decision-feature estimator ranks wrong workflow actions better than the model's own final-step confidence. The intervals overlap only for OLMo-2.

amendments are disclosed in Appendix D, including a readout bug fixed after interim results, with all general models re-run.

All 8 pre-registered tests are supported (Fig. 4 and Table 2). Decision estimators reach AUROC 0.64–0.76 and final-step confidence 0.40–0.58; the weakest model is OLMo-2 (0.637 vs. 0.551; one-sided $p = 0.037$, passing at the last Holm step). Jev's frozen estimator, never refitted, reaches 0.76 [0.72, 0.80] on OpenJev's outputs, as well as OpenJev's own estimator does. The E_{dec} intervals of Jev, OpenJev and Qwen2.5 overlap, so the models cannot be ranked against each other; only the within-model contrast is the finding.

The signal is simple. A class-frequency rule, the historical error rate of the action the workflow chose, estimated on development bugs only, matches the estimator for Jev (0.76) and OpenJev (0.75) and comes close for the others (0.58–0.70). The predictive information is mostly *which action was chosen* and how often that action has been wrong. Part of this is structural: the action and the target are both outputs of the same rule F, so an action's error rate is a property of the workflow as much as of the model. We therefore do not read the result as the estimator being sophisticated, but as evidence that per-step confidence misses information the workflow itself already exposes. Confidence is not useless, though: adding the uncertainty features to the decision features raises held-out AUROC by +0.006 (Jev), +0.004 (OpenJev) and +0.012 (Qwen2.5), but by +0.054 (Phi-3.5) and +0.078 (OLMo-2). These are point estimates without paired intervals. For the Jev-style models confidence is largely redundant with the decisions; for the weaker general models raw token probabilities carry some extra information.

			Held-out AUROC for wrong actions				Cost at 10% review budget			
Model	Type	Fail rate	Decision est.	Final-step conf.	Action rule	Jev est.	No review	Best review arm	Oracle	Wrong accepts
Jev	TypeSafe, hosted	0.43	0.76 [0.72, 0.80]	0.55 [0.49, 0.60]	0.76	—	658	681 (conf. product)	370	52%
OpenJev	Jev-style, 27B	0.47	0.75 [0.70, 0.79]	0.57 [0.53, 0.63]	0.75	0.76	660	674 (confidence)	441	40%
Qwen2.5	general, 7B	0.74	0.74 [0.68, 0.79]	0.47 [0.41, 0.54]	0.70	0.74	903	910 (estimator)	856	14%
Phi-3.5	general, 3.8B	0.72	0.68 [0.62, 0.73]	0.40 [0.34, 0.46]	0.69	0.70	831	890 (confidence)	890	4%
OLMo-2	general, 7B	0.80	0.64 [0.58, 0.69]	0.55 [0.48, 0.62]	0.58	0.64	893	916 (all arms tie)	916	1%

Table 2: Per-model results on the same 500 held-out bugs. AUROC columns rank wrong workflow actions (95% bootstrap CI): the decision-feature estimator fitted on each model's own development outputs, the model's final-step confidence, the action-conditional error-rate rule, and Jev's frozen estimator applied unchanged. Cost columns are from the pre-registered intervention test (Section 7); the oracle is an upper bound that must spend all 50 reviews, and "wrong accepts" is their share of no-review cost. Absolute failure rates are not a clean quality comparison, because the Jev-style and general models use different interfaces.

Model	A1 − random	A1 − confidence
Jev	-9.8 [-17.4, -2.6]*	-4.3 [-39.6, $+36.9$]
OpenJev	-12.2 [-22.2, -3.6]*	$+14.5$ [-30.9, $+55.2$]
Qwen2.5	$+2.3$ [-0.8, $+7.0$]	$+1.5$ [-28.1, $+39.2$]
Phi-3.5	-0.1 [-1.2, $+0.8$]	$+8.4$ [$+0.0$, $+22.7$]
OLMo-2	$+0.0$ [$+0.0$, $+0.0$]	$+0.0$ [$+0.0$, $+0.0$]

Table 3: Pre-registered intervention tests: difference in total cost (A1 minus comparator; negative favors A1) at a 10% budget, with 95% bootstrap intervals. *Supported after Holm correction across all 10 tests.

m	h	p^*	Jev	OpenJev	Qwen2.5	Phi-3.5	OLMo-2
0.5	0.5	25%	+29	+25	+21	+52	+20
0.5	1	33%	+54	+50	+46	+77	+28
0.5	2	50%	+104	+100	+96	+127	+45
1	0.5	14%	**-2**	**-11**	**-18**	+34	+14
1	1†	18%	+23	+14	+7	+59	+23
1	2	27%	+73	+64	+57	+109	+40
2	0.5	7%	**-69**	**-83**	**-97**	**-2**	+4
2	1	10%	**-44**	**-58**	**-72**	+23	+13
2	2	14%	+6	**-8**	**-22**	+73	+30

Table 4: Exploratory cost sensitivity: best feasible arm minus no review (negative, in bold: review pays), with rankings frozen as pre-registered. m scales wrong-accept costs, h is the human cost per escalation, and p^* is the break-even precision for reviewing an accepted case whose target is *escalate*. †The pre-registered table. Point estimates only.

Failure modes differ by model family. Jev and OpenJev both pull severity to the middle level. The general instruct models push severity to "blocking" for 67–100% of bugs whatever the true level, and are wrong far more often (72–80% vs. 43–47%).

7 Prediction is not intervention utility

Design. Before computing any cost outcome, we pre-registered a review test on the held-out bugs: 50 reviews (10%) per model, each escalating a case to a human and priced by the frozen matrix. The primary arm (A1) ranks candidates by the *action-level expected saving* of a review, estimated on development bugs; it is cost-aware rather than error-rate-based, because reviewing a wrong *defer* never saves anything. Comparators: confidence-ranked review (B) and random review (C); Holm across 10 tests.

Result. Two of ten pre-registered tests are supported (Table 3): A1 beats random for Jev and OpenJev, by about 10–12 cost units. A1 is not shown to beat confidence for any model (for Jev its point estimate is 4 units lower, with an interval spanning zero). Every review arm costs more than no review at all; for Jev, 658 with no review against 681–714 with review (Fig. 5).

Sensitivity to costs (exploratory). The pre-registered result holds under one frozen cost table. In a post-hoc analysis planned before computing it, we re-scored the same frozen rankings on a grid of cost ratios within a factor of 2 of that table (Table 4). The conclusion is cost-robust only for OLMo-2. For Jev, OpenJev and Qwen2.5, review pays when wrong accepts cost twice as much (for Jev, 44–69 units below no review) or when review costs half as much, because the break-even precision p^* then falls to about the 14–16% share of wrongly accepted cases. Some reversals are small (Jev -2 at $m = 1$, $h = 0.5$) and lie well inside the intervals of the original tests. The ordering is more stable than the sign: the action rule beats random for both Jev-style models in 12 of 16 cells. The stopping decision below was made under the frozen table and, by its rule, is not reopened.

Where the cost is. For the Jev-style models, 40–52% of no-review cost comes from wrongly *accepted* serious or regression bugs. An oracle that picked the right 50 cases would save 33–44%. But those cases are 15–16% of accepted cases.

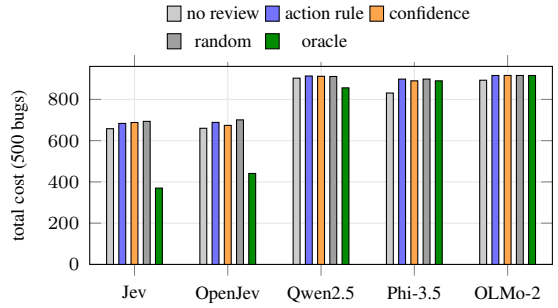


Figure 5: Total cost at a 10% review budget. No feasible strategy beats no review. A perfect reviewer (oracle, an upper bound using held-out outcomes) would cut Jev's cost by 44% and OpenJev's by 33%; for the general models, which escalate most cases themselves, it saves little or nothing.

Fresh bugs	Budget	k	Power	Median Δcost	80% range
500	1%	5	0.13	-1	$[-23, +10]$
	2%	10	0.07	-2	$[-24, +20]$
	3%	15	0.05	$+8$	$[-25, +30]$
	5%	25	0.03	$+17$	$[-16, +42]$
	7.5%	38	0.03	$+24$	$[-23, +65]$
	10%	50	0.01	$+26$	$[-32, +78]$
750	1%	8	0.12	-6	$[-39, +16]$
	2%	15	0.10	$+8$	$[-36, +30]$
	3%	23	0.03	$+13$	$[-31, +46]$
	5%	38	0.04	$+32$	$[-23, +65]$
	7.5%	57	0.02	$+43$	$[-29, +92]$
	10%	75	0.01	$+43$	$[-26, +106]$
1,000	1%	10	0.20	-2	$[-46, +20]$
	2%	20	0.14	$+7$	$[-59, +40]$
	3%	30	0.08	$+16$	$[-50, +60]$
	5%	50	0.03	$+38$	$[-32, +100]$
	7.5%	75	0.03	$+57$	$[-39, +120]$
	10%	100	0.04	$+52$	$[-52, +145]$

Table 5: Pre-registered feasibility check for the case-level score: simulated power to show lower cost than no review (one-sided $\alpha = 0.05$), and the median and 80% range of the projected cost change, over 400 simulated studies per cell under the natural label mix. Viability required power ≥ 0.80; no cell met it.

The action rule cannot separate them, because every accept looks alike to it, and confidence cannot either. The value exists; the identification fails. For the general models, cost is dominated by over-escalation, which more review cannot reduce. Ranking which actions are wrong is therefore not the same as finding which reviews save money: frequent errors and expensive errors differ, the distinction Park et al. (2026) draw in general.

A case-level score, and a stopping decision. We then asked whether a per-case score could find the expensive errors. Under a plan committed before any score was computed, a parameter-free plug-in score was selected on the 442 development bugs over a small logistic model and frozen. It ranks accepted cases by expected saving from review, $9p_2 + 6p_1q - 2(1 - p_2 - p_1q)$, using `score`'s mass on levels 1 and 2 (p_1, p_2) and `noul`'s probability q. On the development mix it cut cost by 12% against no review (630 vs. 713) and captured 22% of the oracle saving. But the development bugs were quota-sampled; reweighted to the natural label mix, a 10% budget was projected to cost *more* than no review. Before generating any fresh case, we pre-registered a feasibility check that varied only the budget (1–10%), with a binding stop rule: viability required power ≥ 0.80 to show lower cost than no review with at most 1,000 fresh bugs, and a negative median projected cost change, with uncertainty in error prevalence and ranking propagated by resampling. No budget was viable. The highest power was 0.20 (1% budget, 1,000 bugs), with a median change of -2 and an 80% range of $[-46, +20]$; every budget of 2% or more had a positive median cost change (Table 5). The confirmatory study was therefore not run, no further designs were searched, and no fresh API calls were made. This is a feasibility assessment, not evidence that review cannot help: a good selector would save substantially, but the signals available here cannot identify the cases well enough to pay for the reviews.

8 Discussion

Relation to prior work. The single-request audits (Deußer et al., 2026; Meng, 2026) establish that Jev's scores must be validated per task and question form; our primitive-level results add that, once chained, the only discriminative confidence can sit on a node the action never reads, and that `score`'s confidence is silent about its own systematic bias. Park et al. (2026) show that risk ranking and review-value ranking differ when errors differ in repairability. Our setting isolates a different source of the same gap, cost asymmetry *within* an action: wrongly accepted serious bugs are expensive, but they share the action, the error history and largely the confidence of the correctly accepted ones, so no action-level or confidence-level signal can separate them.

Lessons for practitioners. First, a per-step confidence is not a workflow-level risk signal; the workflow's own action and that action's error history predicted failures better for every model we evaluated. Second, a better failure ranking should

not be deployed as a review policy without scoring it in cost: here the most error-prone action was a cheap one, ranking by error rate spent reviews where they saved nothing, and whether any review paid depended on the ratio of wrong-accept cost to review cost. Third, where review value exists it may sit in errors that no available signal separates; a pre-registered feasibility check can establish this before data are spent. Simple process checks also caught most of our own design defects before data existed (Appendix E).

What would make review pay. The oracle bounds the prize at 33–44% of cost for the Jev-style models. Capturing it needs a signal that separates serious from minor bugs *among accepted cases*, for example evidence the workflow does not currently read, or a primitive whose severity confidence tracks severity. Whether such a signal exists is an open question that our data cannot answer.

9 Conclusion

What a typed-decision workflow *decided* predicted its failures better than how confident it was, on 500 new bugs and across five models. Under the pre-registered costs, that prediction did not become a cheaper review policy, and whether review pays at all depends on the cost ratio: the expensive errors are real, but they hide inside actions that look alike, and a case-level score that targeted them failed a pre-registered feasibility check, so we stopped by rule. Workflow uncertainty has to be evaluated against the decision it is meant to improve.

Limitations

Five models in one domain (bug triage) and one workflow; Jev was version 1.13.0, and later versions or other domains may behave differently. The general instruct models use our prompt and readout, not a trained decision interface, so their absolute failure rates are not a clean comparison. Reference labels are not ground truth: Mozilla's sample excludes re-triaged bugs, and kernel severity is the reporter's claim with "normal" as a default. The target action is a construct we defined, and all strategy results are conditional on it. Old public bugs may appear in the models' training data. The cross-model study has three disclosed amendments, including a readout bug fixed after interim results were seen. The intervention test used one budget and one cost table, both frozen in advance; the cost-sensitivity analysis is exploratory, uses point estimates, and real operational costs are unknown; reviewers are assumed correct, and the oracle is an upper bound, not a strategy. The feasibility check is a simulation on development data and is not confirmatory evidence in either direction. The protocol went through twelve dated revisions, each made before the data it affects existed.

Ethical considerations

All bug reports are public; texts were privacy-scrubbed, and the released data will contain identifiers and hashes rather than report text. No human subjects were involved. API and GPU costs were modest. AI assistants were used for code, analysis scripts and drafting; the authors reviewed all claims, and every reported number is regenerated from committed results by a traceability script.

References

Marah Abdin, Jyoti Aneja, Hany Awadalla, et al. 2024. Phi-3 technical report: A highly capable language model locally on your phone. *arXiv preprint arXiv:2404.14219*.

Jean V. Alves, Diogo Leitão, Sérgio Jesus, Marco O. P. Sampaio, Javier Liébana, Pedro Saleiro, Mário A. T. Figueiredo, and Pedro Bizarro. 2024. Cost-sensitive learning to defer to multiple experts with workload constraints. *Transactions on Machine Learning Research*.

Dylan Bouchard. 2026. Is escalation worth it? On the depth of LLM cascades. *arXiv preprint arXiv:2605.06350*. V2; v1 was titled "Is Escalation Worth It? A Decision-Theoretic Characterization of LLM Cascades".

Aritra Chowdhury, Malik Magdon-Ismail, and Bülent Yener. 2019. Quantifying contribution and propagation of error from computational steps, algorithms and hyperparameter choices in image classification pipelines. *arXiv preprint arXiv:1903.00405*.

Steven Denney and Matthew DiGiuseppe. 2026. JEV versus LLMs: Accuracy, cost and calibration on seven political science replications. *arXiv preprint arXiv:2610.06625*.

Tobias Deußer, Lorenz Sparrenberg, and Rafet Sifa. 2026. Evaluating and benchmarking the system one model Jev. *arXiv preprint arXiv:2609.37647*.

Yonatan Geifman and Ran El-Yaniv. 2017. Selective classification for deep neural networks. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Jonathan Geuter, Andrew Cheng, and David Alvarez-Melis. 2026. Releasing the inner Jev: A simple recipe for open decision models. *arXiv preprint arXiv:2610.17205*.

Jaehun Jung, Faeze Brahman, and Yejin Choi. 2025. Trust or escalate: LLM judges with provable guarantees for human agreement. In *International Conference on Learning Representations (ICLR)*.

Josef Kittler, Mohamad Hatef, Robert P. W. Duin, and Jiri Matas. 1998. On combining classifiers. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(3):226–239.

Varun Kotte. 2026. PASC: Pipeline-aware conformal prediction with joint coverage guarantees for multi-stage NLP and LLM pipelines. *arXiv preprint arXiv:2605.18812*.

Moule Lin, Qizhen Lan, Shuhao Guan, Weipeng Jing, Jiexin Fan, David Gregg, and Goetz Botterweck. 2026. Uncertainty quantification for LLM agents: A taxonomy, an evaluation protocol, and an empirical study. *arXiv preprint arXiv:2609.07395*.

David Madras, Toniann Pitassi, and Richard Zemel. 2018. Predict responsibly: Improving fairness and accuracy by learning to defer. In *Advances in Neural Information Processing Systems (NeurIPS)*.

Lingsen Meng. 2026. How far can a commercial decision model's probabilities be trusted? A calibration audit of TypeSafe Jev against open and general-purpose classifiers. Code, data and paper, Zenodo, https://doi.org/10.5281/zenodo.22935043. Version 1.0.0.

OpenJev. 2026. OpenJev: an open-weights decision model. https://huggingface.co/openjev/openjev. Hugging Face commit b4a89f5; CC BY-NC 4.0 weights; states it is independent of TypeSafe. Not the same project as other repositories named OpenJev. Accessed 2026-10-17.

SangJin Park, Myungsub Choi, Jineok Kim, and Minseung Kang. 2026. Risk is not review value: Wrong-answer exposure under bounded review budgets. In *SeT-LLM Workshop at KDD*. ArXiv:2609.07095.

Qwen, An Yang, et al. 2024. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*.

Lijuan Tang and Yuemeng Zheng. 2026. Typed decision models: An early evidence audit and evaluation checklist. *arXiv preprint arXiv:2609.32160*.

Team OLMo, Pete Walsh, Luca Soldaini, et al. 2024. 2 OLMo 2 furious. *arXiv preprint arXiv:2501.00656*.

Jiaqing Zhang, Sandeep Elluri, Bhanu Cherukuvada, Yonah Joffe, Jessica Sena, Miguel Contreras, Scott Siegel, Subhash Nerella, Catherine Price, and Parisa Rashidi. 2026. Auditing multimodal LLM raters: Central tendency bias in clinical ordinal scoring. *arXiv preprint arXiv:2605.16386*.

Table 6: The action rule $F(c_2, c_3)$. *Unclear* is checked first.

$c_2 \setminus c_3$	0	1	2
1 (yes)	accept	defer	escalate
0 (no)	accept	accept	escalate
unclear	request evidence (any c_3)		

Table 7: Frozen cost matrix $\mathrm{cost}(a, a^*)$. One unit of human cost is added per escalation. "req. ev." = request evidence.

action \ target	accept	escalate	req. ev.	defer
accept	0	10	6	8
escalate	1	0	1	1
req. ev.	1	1	0	1
defer	1	1	1	0

A Action rule and cost matrix

B The primitives: full results

Severity labels: Mozilla S1–S4 map to levels 2/2/1/0 (triage); kernel blocking/high/normal/low map to 2/2/1/0 (reporter). With Mozilla's own severity definitions in place of ours, agreement moves only from 0.53 to 0.56, so the compression persists. `noul`'s errors change direction between trackers: on Mozilla it mostly misses regressions (23 misses vs. 17 false alarms); on the kernel it over-flags them (63 false alarms vs. 3 misses).

C Exploratory composition analyses

All analyses here were specified after the Mozilla results had been seen and are hypothesis-generating.

The naive comparison is confounded. On Mozilla, decision composition (S1) beats confidence composition (S2) by 0.17 [0.08, 0.27] in $P(Y_{\text{decision}})$, with 61 vs. 27 discordant pairs (exact McNemar $p = 0.0004$). But S2 never reads the node decisions, while a^* escalates on the severity *decision*. To be right on a serious bug, S2 needs low confidence to *mean* high severity, so the comparison mostly measures whether confidence can stand in for severity. On the kernel the difference is smaller and inconclusive: 0.06 $[-0.03, 0.15]$, 66 vs. 51 discordant pairs, $p = 0.20$. The difference between trackers is 0.11 $[-0.01, 0.24]$. Its interval covers zero, so we have *no evidence that the effect differs* between trackers, only a weaker estimate on one of them.

Table 8: Per-primitive behavior of the confidence signal. AUROC is for predicting the node's own agreement with the reference label, with 95% bootstrap intervals.

	Constructed items	Mozilla (n=202, triage)	Kernel (n=240, reporter)
`choice`	High-confidence error subgroup: mean confidence 0.915, accuracy 0.333 (n=6)	85% at $\geq$ 0.90, agreeing 0.96; AUROC 0.90 [0.83, 0.96]; ECE 0.05	96% at $\geq$ 0.90, agreeing 0.99; AUROC 0.98 [0.95, 1.00]; ECE 0.02
`noul`	24/24 correct, confidence below accuracy in every bin	Agreement flat across bands (0.71–0.84); AUROC 0.53 [0.42, 0.65]; ECE 0.36	Non-monotonic (0.58–0.93); AUROC 0.41 [0.32, 0.50]; ECE 0.36
`score`	Tracks the most severe sentence (6/6); confidence varies by 0.173 with sentence order alone (1 of 3 items)	Confidence ≈0.6 at every true level; AUROC 0.58 [0.50, 0.66]	Confidence 0.65 / 0.62 / 0.61 by true level; AUROC 0.57 [0.50, 0.64]

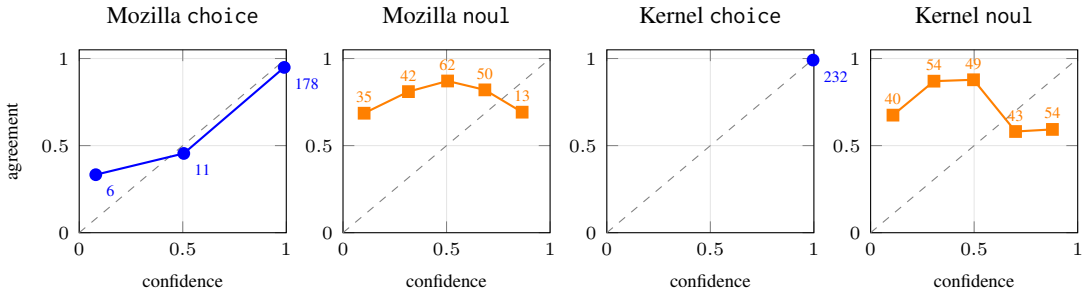


Figure 6: Reliability diagrams (five equal-width confidence bins, bins with $n \geq 5$; the label is n; dashed line: perfect calibration). `choice` puts nearly all bugs in its top bin, and that bin is accurate. `noul`'s confidence-like value $2|\bar{p} - 0.5|$ sits far *below* its agreement (underconfidence), and agreement does not rise with it.

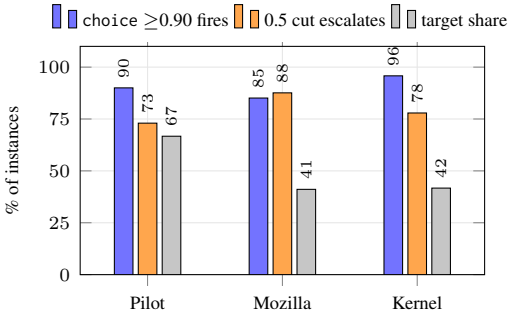


Figure 7: Selectivity of the two fixed-threshold rules, set on constructed data and measured on each rule's own input signals. The trigger fires on 85–96% of instances everywhere. The product cut is near its target on the constructed pilot and far above it on real text.

Budgeted overrides, scored by cost. At a fixed budget of $k = \lceil 0.1n \rceil$ overrides ($k = 21$ on Mozilla, 24 on the kernel), let T be the total cost of the confidence-ranked overrides and μ the exact expected total under k overrides drawn uniformly from the same candidates. Two analyses answer two different questions.

- **Randomization test, conditional on the evaluated bugs.** This asks whether the ranking selected worse or better overrides than random selection *on these bugs*. The statistic is $|T - \mu|$. The null distribution comes from 10,000 random k-subsets of the candidates, and the test is two-sided.

- **Bootstrap over bugs.** This asks how stable the per-bug difference $(T - \mu)/n$ is under resampling of the bugs, with rankings and μ recomputed in each resample.

On Mozilla, $T = 390$ against $\mu = 357.2$. The observed value sits at the edge of the null distribution (2.5–97.5% range 323–388), so $p = 0.05$. The per-bug difference is $+0.16$ $[-0.01, 0.24]$, and 97% of resamples are positive. That is borderline evidence that the product ranking chose *worse* overrides than random on Mozilla. On the kernel, $T = 338$ against $\mu = 358.6$ ($p = 0.18$), a per-bug difference of -0.09 $[-0.23, 0.06]$: inconclusive, and in the opposite direction. These analyses are exploratory and uncorrected for multiplicity. We read them as no consistent benefit from confidence-ranked overrides, with one borderline harmful estimate that did not replicate, and not as an established harm.

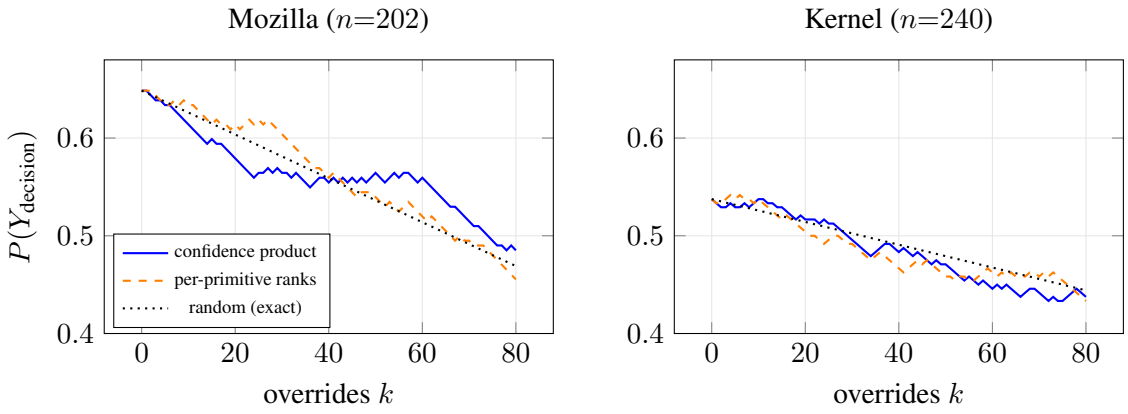


Figure 8: Adding confidence-ranked escalations on top of decision composition. At $k = 0$ the curve is S1. Neither ranking consistently improves on random selection of the same number of cases: deviations are a few points at most and change sign between rankings and trackers. For example, per-primitive ranks run slightly above random on Mozilla for $k \approx 15$–35, and the confidence product runs below.

A structural reason. In this workflow the only discriminative confidence (`choice`, AUROC 0.90–0.98) belongs to the one node that F does not use. The informative signal never reaches the action, and the two signals that do reach it are close to uninformative.

D Pre-registration deviations

- **Cross-model, amendment 1:** OpenJev was added before any of its outputs existed.
- **Amendment 2:** OLMo-2 needed a newer `transformers` release.
- **Amendment 3:** a readout bug, in which a SentencePiece word-start token was shared by all digit labels, was found after interim results had been seen. All three general models were re-run under one corrected readout; the conclusions did not change. On 3% of OLMo-2's requests most probability fell outside the allowed labels; excluding them does not change its result.
- **Protocol revisions:** the development protocol went through twelve dated revisions, each made before the data it affects existed.

E Audits that caught defects

Table 9 lists defects the study's own audits caught, most before the data that would have exposed them existed. We recommend asking, for every strategy and target class, *can this strategy be right here, and through what information?*, and measuring each trigger's selectivity on pilot inputs before freezing an escalation policy.

F Reproducibility

All calls used `jev-1.13.0`: 4,584 in the development runs and 4,500 on the held-out bugs; each open model served 2,826 requests. Every dataset, raw response, manifest and analysis is hash-locked, as are the pre-registration documents and their revision history. An anonymized code and data archive accompanies the submission; report texts are replaced by identifiers and hashes and can be regenerated from the public trackers.

Table 9: Defects found by the study's own audits.

Defect	Detected by	When
A "primitive-aware" arm structurally identical to the baseline	inspecting the action rule	before evaluation
A trigger firing on ≈90% of inputs	selectivity on input signals only	before evaluation
A threshold sensitive to the data distribution	smoke test vs. pilot	before the full run
Four silent runner defects (e.g. the literal "None" shown to the model; partial data analyzed)	adversarial probing	before the live run
The confidence-vs-decisions confound	post-hoc review	after the Mozilla result
Template content whose labels were recoverable from punctuation alone (100%) and whose sentences leaked cross-label cues	independent review (separate AI agent)	before the controlled study